

Extraction of Lipids from Foods

Laboratory Report — Soxhlet / Solvent Extraction Method

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Technique category: Food Science / Analytical Chemistry

1. Objective

To determine the total fat content of four food samples using solvent extraction, and to classify each sample by fat category based on percent fat (dry weight basis).

2. Introduction

Total lipid content in food samples is commonly determined using solvent extraction methods such as Soxhlet extraction, in which an organic solvent (typically petroleum ether or hexane) continuously washes over a dried, ground food sample, dissolving and extracting non-polar lipid components over repeated reflux cycles. After extraction, the solvent is evaporated off, leaving the extracted fat, which is weighed to calculate the percentage of fat in the original sample.

3. Materials

- Dried, ground food samples (peanuts, oats, spinach, cheddar cheese)
- Petroleum ether or hexane (extraction solvent)
- Soxhlet extraction apparatus
- Analytical balance and drying oven
- Extraction thimbles (cellulose)

4. Method

- Dried each food sample to constant weight (~100–105°C) and recorded dry weight.
- Ground samples finely and loaded into pre-weighed extraction thimbles.
- Loaded thimbles into the Soxhlet apparatus and refluxed with solvent for 4–8 hours.
- Evaporated residual solvent from the extraction flask.
- Weighed the extracted fat by difference and calculated percent fat relative to dry sample weight.

5. Results

Fat content varied substantially across the four samples, ranging from 1.6% in dried spinach to 48.0% in peanuts.

Sample	Dry Wt (g)	Fat Extracted (g)	% Fat	Category
Peanuts	5.02	2.41	48.0	High-fat
Oats	5.10	0.34	6.7	Moderate-fat

Spinach (dry)	5.00	0.08	1.6	Low-fat
Cheddar	5.05	1.68	33.3	High-fat

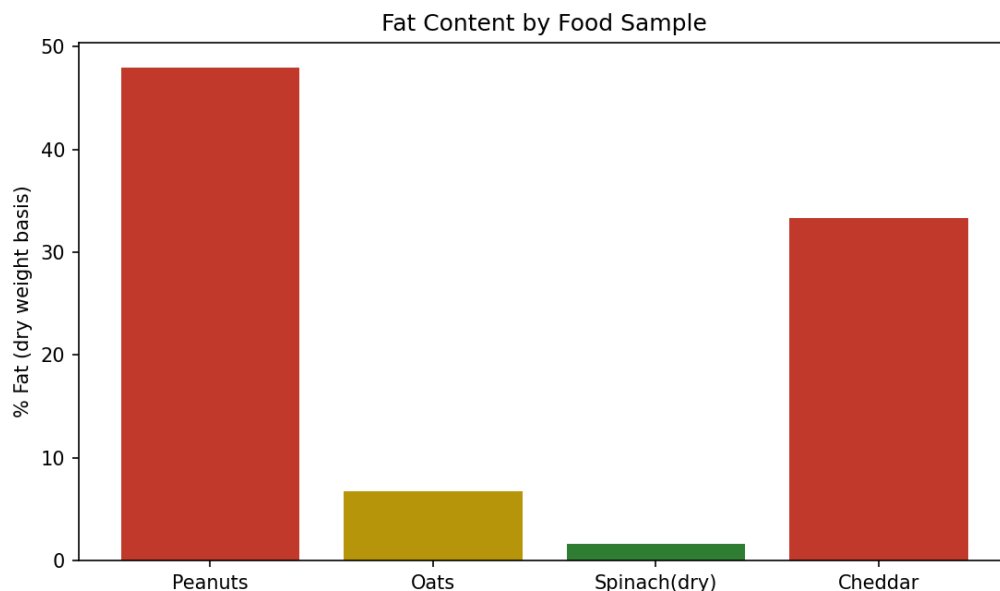


Figure 1. Percent fat content (dry weight basis) by food sample, coloured by fat category.

6. Discussion

Peanuts and cheddar cheese, both classified as high-fat, showed fat contents consistent with their known nutritional profiles — peanuts are a nut with a naturally high oil content, while cheddar is a full-fat dairy product. Oats, classified as moderate-fat, reflects the modest but non-negligible lipid content typical of whole grains. Dried spinach showed the lowest fat content by a wide margin, consistent with leafy vegetables being predominantly composed of water (removed during drying), fibre, and carbohydrate rather than lipid. These results demonstrate the wide range of fat content across food categories that solvent extraction is routinely used to quantify for nutritional labelling and food science applications.

7. Conclusion

This exercise demonstrated the complete solvent extraction workflow for quantifying food fat content, from sample drying through extraction to gravimetric calculation, and confirmed expected fat content patterns across a diverse set of food types.